

120 Days of Quantitative Finance

Week 16: Stochastic Calculus + Time Series

Day 107: Linking SDEs with ARIMA and GARCH Models

1 Introduction

Financial modeling can be approached from two perspectives:

- Continuous-time models (SDEs)
- Discrete-time models (ARIMA, GARCH)

This section establishes the connection between these frameworks.

2 Continuous-Time Model

Asset prices are often modeled using SDEs:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

This represents continuous evolution with randomness.

3 Discrete-Time Approximation

Using Euler discretization:

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sigma S_t \sqrt{\Delta t} \epsilon_t$$

$$\epsilon_t \sim \mathcal{N}(0, 1)$$

4 Log Returns Representation

Taking logarithms:

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right)$$

This leads to:

$$r_t = \mu \Delta t + \sigma \sqrt{\Delta t} \epsilon_t$$

5 Link to ARIMA Models

In discrete time:

$$r_t = c + \phi r_{t-1} + \epsilon_t$$

ARIMA models approximate continuous dynamics through lagged dependencies.

6 Link to GARCH Models

Instead of constant volatility:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

This introduces time-varying volatility.

7 Key Insight

- SDEs assume continuous-time stochastic processes
- ARIMA approximates drift dynamics
- GARCH approximates volatility dynamics

8 Unified Representation

$$r_t = \mu_t + \sigma_t \epsilon_t$$

where:

- μ_t is modeled by ARIMA
- σ_t is modeled by GARCH

9 Financial Interpretation

- Continuous-time models are used for pricing derivatives
- Discrete-time models are used for estimation and forecasting

10 Conclusion

SDEs and ARIMA/GARCH models are not competing frameworks but complementary tools.

Together, they provide a complete view of financial dynamics.